Financial Institutions & Markets- ENT 309

PPT #4, IIITD, Summer, 2026

Agenda

- Duration & Modified Duration

- Types of Bonds
- Innovations in Bonds
- Bond Yields (Types)
- Different types of risks in Bonds
- Bonds Ratings

Duration- D

Weight associated with the cash flow made at time t(denoted by CFt)

$$w_t = \frac{\mathrm{CF}_t/(1+y)^t}{\text{Bond price}}$$

- CF -> Cash Flow
- y -> YTM/ Discount Rate
- t -> time period

Macaulay's Duration (we will refer to it as Duration for practical reasons)

$$D = \sum_{t=1}^{T} t \times w_t$$

- DURATION IS A MEASURE OF THE "EFFECTIVE MATURITY" OF THE BOND.
- Another useful concept is Modified Duration.

Modified Duration-D^*

- We are interested in Change in Bond Price which depends on duration
- Modified Duration (D^*) helps estimate change in bond price as YTM changes. Sensitivity of Bond Price to Interest Rates.
- $D^* = (-)(1/P)(dP/dy) = D/(1+y)$ where P is the bond price, D is Macaulay Duration and y is the interest rate

OR (for small changes in y), the % change in price is

$$\frac{\Delta P}{P} = -D^* \Delta y \quad \textbf{where} \quad D^* = \frac{D}{1+y}$$

- Unit of D & D^* is same as unit of time. Example: years.

Modified Duration-D^*

Example:

- Bond 5/6 have a modified duration of 4.08/6.14 years.
- Note that we can calculate the approximate percentage change in price of a bond as:

$$\% \Delta P = -D^* \times \Delta y$$

So, when rates drop by 2%, we would find that Bond 5/6 should change by approximately:

$$\%\Delta P_{Bond5} = -4.08 \times -0.02 = 0.0816 = 8.16\%$$

$$\%\Delta P_{Bond6} = -6.14 \times -0.02 = 0.1229 = 12.29\%$$

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Types of Bonds

Types of Bonds

Issuer
Corporation
Government
International

Coupon Rate
Fixed Income
Floater
Inverse Floater
Zero Coupon

Redemption Features
Callable
Convertible
Puttable

Priority
Junior or Subordinated
Senior or Unsubordinated

Govt Securities
Bonds (10 yr+)
Notes (1-10 yrs)
T-Bills (<1 yr.)

International Bond Issues
Eurobond
Foreign

Types of Bonds

<u>Coupon Rate</u>
Fixed Income
Floater
Inverse Floater
Zero Coupon

- Fixed income: Fixed coupon.
- Floater: Linked to a benchmark interest rate. Repo Rate, T-bills/ 10 yr bond yield. Coupon goes up and down with benchmark movement.
- Inverse Floater: Coupon rate moves in the direction opposite to benchmark.
- Zero coupon: As discussed earlier

Types of Bonds

Redemption Features

Callable
Convertible
Puttable

- Features of Corporate Bonds
- Callable: May be repurchased by **issuer** at specified call price during call period
    - Callable bonds should offer a lower or higher coupon. Why?
- Convertible bonds: Allow bondholder to exchange bond for specified number of common stock shares
- Puttable bonds: **Holder** may choose to exchange for par value or to extend for given number of years. On the **call date.**

Types of Bonds

<u>International Bond Issues</u>

Eurobond

Foreign

- International bond issues are debt securities sold largely outside the country of residence of the borrower often in many countries simultaneously. Sub divided in Eurobonds and Foreign bonds.
- Bonds are <u>underwritten by</u> an international syndicate of commercial and investment banks.
- Eurobonds: Sold by an issuer in other national markets and denominated in a currency which is not issuer's or of that national market.
- HDFC Bank selling USD bonds in Europe- Eurodollar bond
- A German company wanting to raise Yen in India will sell bond in India in Yen- Euroyen bond

Types of Bonds

International Bond Issues

Eurobond

Foreign

- **Foreign bonds:** Bonds are underwritten by a syndicate composed of commercial and investment banks from **one country**:
- Foreign Bonds are Denominated in that country's currency
- Sold principally in that country.
  - Yankee Bond - A bond denominated in U.S. dollars that is publicly issued in the U.S. by foreign banks and corporations
  - Bulldog Bond - A sterling denominated bond that is issued in London by a company that is not British
  - Kangaroo Bond – Aus Dollar denominated bond. Australian market. By a foreign entity
  - Samurai Bond – Yen denominated bond. Japanese market.. By a non-Japanese co.

Types of Bonds

International Bond Issues

Masala Bond: Inspired by Eurobond

- The Process: The International Finance Corporation (IFC) issued rupee-denominated bonds worth ₹1,000 crores on the London Stock Exchange.
- How Investors Participate: A US or UK investor converts their Dollars or Pounds into Rupees to buy the bond at the prevailing exchange rate.
- The Return: At the end of the bond's term, the IFC pays back the principal plus interest in Rupees. The investor must then convert those Rupees back into their native currency.
- What will you call such a bond?
- Later HDFC & NTPC also issued such bonds. Help guard the bond issuer against FX risk.

Types of Bonds:Innovations

- Inverse Floater
- Asset backed bonds
- Pay in Kind Bonds
- Catastrophe Bonds
- Index Bonds
- Treasury Inflation Protected Securities (TIPS)

Types of Bonds:Innovations

- Inverse Floater: Coupon change is 'inverse' to change in Benchmark rates.
  - How does increase in benchmark rate impact Inverse Floater bonds?
  - Coupon Payouts reduce. Discounting factor increases, P falls. Double impact.
- Asset Backed Securities: Income from specific assets to service coupon payments. Linked to a <u>distributed portfolio of assets</u>.
  - Mortgage backed securities. Bonds backed by production from oil wells.
  - Secured bonds typically have <u>A SPECIFIC ASSET</u> of the issuer as a collateral.
  - Unsecured bonds are not linked to any collateral.
- Pay in kind: Choice with the issuer. In case he does not have money to pay interest due to cash crunch can pay in 'more bonds'.

Types of Bonds:Innovations

- Catastrophe Bonds: Principal is not paid if a bad event happens. To compensate higher coupons rates are paid. Ex. FIFA world cup. Event: If terrorism halts the cup.
- Index Bonds: Payments tied to inflation/general price index / price of a particular commodity (ex. oil). Could be
  (a.) Only coupon payments linked to index
  (b.) Coupon payments and Maturity value both linked to index
- TIPS- Treasury Inflation Protected Security: Issued by US Govt. Par Value of the bond increases/decreases with consumer price index.
  - Coupon payments change.
  - And maturity value also changes
  - Redemption is at higher of Face Value or Maturity value.

Bonds Innovations: SGB

- **Sovereign Gold Bonds**: Issued by RBI on behalf of central government.
- Many series issued from 2015 till 2024.
- To replace purchase of physical gold as investment. In multiples of 1 gm.
- Pays a coupon of 2.5% per year. On the amount initially invested. Semi-annual.
- Maturity: 8 years with exit option after 5 years
- Redemption linked to market price of gold. Can be traded in secondary market. Can be used as collateral for loans.
- To reduce import of gold. Save Forex.
- What was the return which initial investors received?
- > 300% cumulative return over 8 years. Compounded Annually at ~ 15%.
- A BIG OOPS!!
    - Government did not hedge SGB payouts/ Gold prices.

https://economictimes.indiatimes.com/markets/bonds/sovereign-gold-bond-one-of-dumbest-government-borrowing-programs-in-world-fundoo-professor/articleshow/123939909.cms?from=mdr

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Types of Bond Yields

Bond Yields: Types

Nominal Yield	• Annual Coupon Rate
Current Yield	• Coupon Rate divided by Current Price
Yield to Maturity	• Discount Rate which makes Sum of Present Value of all bond payments equal to price
Premium Bonds	• Bonds selling above Face Value
Discount Bonds	• Bonds selling below Face Value

Bonds Yields: Types

- A Rs.1000; 10% bond with 2 years to maturity selling in the market at Rs.980.

- Nominal Yield= ?
  - 10%
- Current yield $=$
  - $\frac{Rs.100}{Rs.980}x100 = 10.204\%$
- YTM
  - $980 = \frac{100}{(1+YTM)} + \frac{100+1000}{(1+YTM)^2}$

Bonds Yields: Types

- Yield to Maturity Versus Holding Period Return (HPR)
- **YTM: Yield to maturity** measures average Rate of Return if investment held until bond matures.
- **HPR** is Rate of Return over particular investment period; depends on market price at end of period. Has two components.
    - Coupon payment or interest payment received during the period
    - The gain or loss in capital due to change in bond price.
- What to do with the coupon payments received?
- What do to if HPR is > Maturity period?

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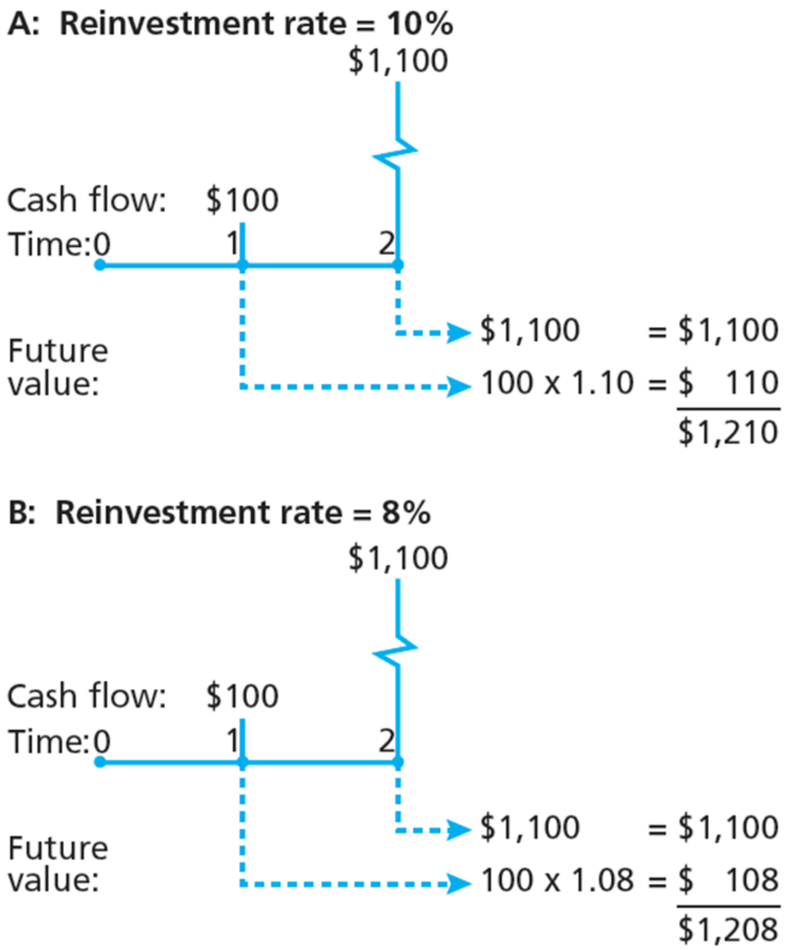
Types of Risks in Bonds & Rating of Bonds

Reinvestment Risk

- The coupon payments might get re-invested at a lower interest rate
- If the bonds matures before the holding period- might have to reinvest at a lower interest rate
- What is the risk if Holding Period < Bond Maturity?
  - Interest rate risk as the bond price might fall if rate rises.



Bonds: Types of Risks

- **Interest Rate Risk**
  - Price Of a bond falls when interest rates rise
  - Reinvestment Risk
  - Floating Rate Securities
- **Credit Risk**
  - Default, Downgrade
- **Contractual Risk**
- **Callable Bonds**
- **Inflation Risk**
  - Increase in inflation – Decrease in Purchasing power

Bond Prices: Credit /Default Risk

- Bond Issuer may default: Unable to pay coupon and/or principal.
- Default risk mainly with corporates. Can countries also default?
- Argentina (2001), Sri Lanka (2022).
- How will the YTM/ discounting factor change for a corporate bond?
- Corporate Bond Yield = Risk Free Yield (Yield on Government bond) + Additional Yield (Credit Spread or Default Premium)
- Credit Spread will depend on the estimated probability of default.
- How to estimate Default Premium?

Bond Prices: Credit /Default Risk

- Investment Grade Bonds
  - Rated BBB and above by Standard & Poor's (S&P) or Baa and above by Moody's
- Speculative Grade or Junk Bonds
  - Rated BB or lower by S&P, Ba or lower by Moody's, or unrated
- +/- or 1,2,3 used to adjust the ratings

Credit Rating Scales by Agency, Long-Term

Moody's	S&P	Fitch	
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC		Extremely speculative
Caa3	CCC-		Default imminent with little prospect for recovery
Ca	CC	CC	
	C	C	
C	D	D	In default
/			
/			

"Junk"

Credit Rating of India

- S & P's: BBB with a Stable outlook. Aug 2025
- Moody's: Baa3 with a Stable outlook. Aug 2023
- Outlook can be Negative, Stable or Positive
- Can the rating of a Corporate Bond be higher than the Country rating?
- Yes. But only for large global corporates whose exposure to home country is limited.
- Reliance (A- by S&P), Infosys Ltd. (A by S&P) etc.

Credit Rating Scales by Agency, Long-Term

Moody's	S&P	Fitch	
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC		Extremely speculative
Caa3	CCC-		Default imminent with little prospect for recovery
Ca	CC	CC	
	C	C	
C	D	D	In default
/			
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Rating Agencies in India

CRISIL A STANDARD & POOR'S COMPANY

ICRA A MOODY'S INVESTORS SERVICE COMPANY

CARE Ratings Professional Risk Opinion

CRISIL	ICRA	CARE	Significance
AAA	LAAA	CARE AAA	Highest safety
AA	LAA	CARE AA	High safety
A	LAA	CARE A	Adequate safety
BBB	LBBB	CARE BBB	Moderate safety
BB	LBB	CARE BB	Inadequate safety
B	LB	CARE B	Risk prone
C	LC	CARE C	Substantial risk
D	LD	CARE D	Default

- & India Ratings and Research (Ind-Ra), Acuite, Brickwork Ratings etc.
- Ratings shopping: What is it?

Price Path: Diff Coupon Rates

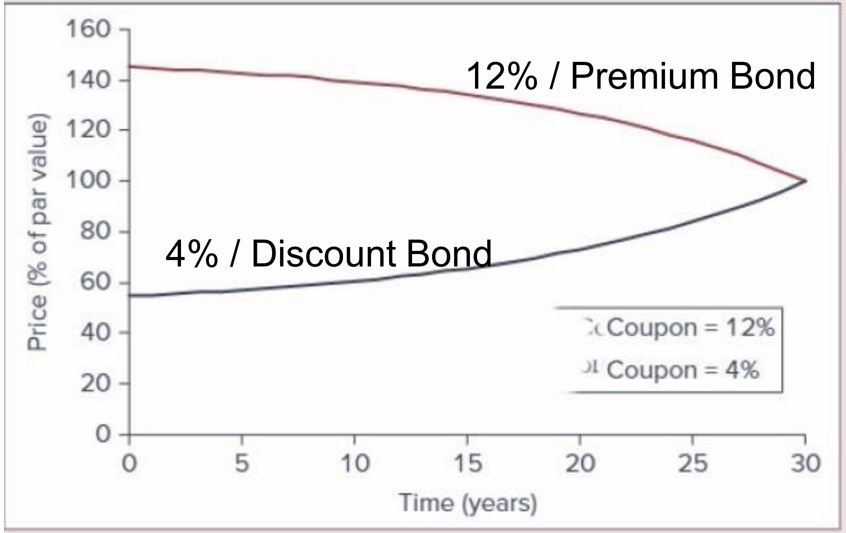


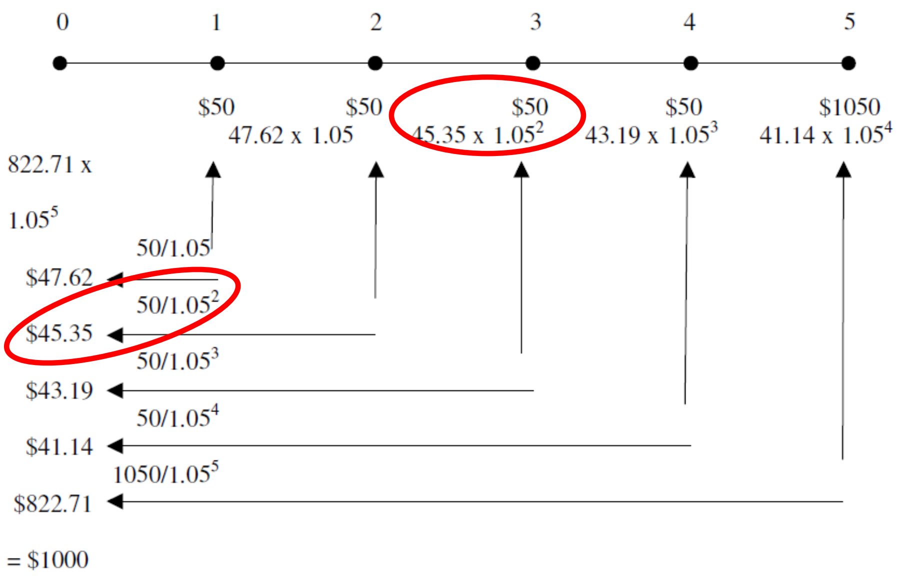
FIGURE 10.6

Price path of two 30-year maturity bonds each selling at a yield to maturity of 8%. Bond price approaches par value as maturity date approaches.

Which one is a PREMIUM bond and which a DISCOUNT bond?

Discounted & Compounded Values

The discounted and compounded values of the coupons and face value of a 5%, 5-year note priced at par to yield 5%



YTM	5%
Maturity	5 years
Face Value & Price	1000

	Principal	Compound Interest	Total	Discounted CF
CF1	47.62	2.4	50	47.62
CF2	45.35	4.6	50	45.35
CF3	43.19	6.8	50	43.19
CF4	41.14	8.9	50	41.14
CF5	822.71	227.3	1050	822.71
TOTAL	1000	250.0	1250.0	1000

- Price of a bond consists of small bundles each of which is <u>compounded</u> at YTM to achieve coupon payments and maturity payment.
- Conversely, each payout is <u>discounted</u> at YTM to determine its contribution to price

Thank You!